

TABLE II

DATASET-USE MATRIX. THE TABLE DISTINGUISHES OFFICIAL EVIDENCE, DEVELOPMENT EVIDENCE, DIAGNOSTIC DATASETS, AND SCAFFOLDING DATA SO THAT TRANSFER CLAIMS ARE NOT INFERRED FROM OPENED OR DERIVED-VIEW RESULTS.

Dataset group	Pairing / modality type	Use in ELARA	Claim ceiling
MVTec 3D-AD	Real RGB + depth/XYZ	Primary naturally paired M1/Gate D benchmark; T5 strong-baseline and controlled-stress replication	Bounded in-domain/stress evidence; not external transfer or one-class SOTA
3D-ADAM	Real RGB + depth	Sealed M2 external transfer and v3 stress-regime transfer; also D13 development evidence	Opened/spent official M2; clean Gate E not confirmed, stress evidence retained
Real-IAD / Real-IAD D3	Real industrial RGB; D3 adds pseudo-3D + point cloud	D13 natural transfer attempt and D16 natural-degradation/headroom audit	D13 fails SAR; D16 all-test CW positive but stress CI crosses zero
MulSen-AD	Real RGB + infrared + point cloud	Development replication for natural positive transfer; strong future fit because modalities are complementary	Development only unless a fresh prelocked split/category is used
Eyecandies / Real3D-AD	Synthetic RGB-D-normal; 3D point-cloud benchmark	Family D failure record and earlier exploratory/mechanism evidence	Useful negative/exploratory evidence; not final transfer confirmation
VisA / MVTec LOCO-AD	RGB with derived edge/noise proxy	Family A/C secondary diagnostics and noise-floor checks	Derived-view proxy only; cannot prove independent multimodal generalization
UNSW-NB15 / healthcare	Structured event or clinical views	Non-vision and clinical-development checks for fusion machinery	Diagnostic/development evidence; not co-observed RGB-X transfer
Fraud / behavior / NLP / vision	Label-aligned tabular/text plus small image scaffold	ELARA-Bench-LA components and utility smoke data	Mechanism/stress/scaffolding only; no natural transfer claim

calibrator when enough labels are available, and computes expected calibration error. At inference time:

$$r_d = \alpha(1 - \text{ECE}_d) + \beta p_{\text{KS},d} + \gamma S_d, \quad (4)$$

where ECE_d is validation calibration error, $p_{\text{KS},d}$ is the Kolmogorov-Smirnov p-value comparing current scores with validation scores, and S_d is score sharpness from squared distance to 0.5. The current benchmark uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

3) *Reliability Gate*: Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, ELARA uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d})r_{i,d}. \quad (5)$$

If the same rule is written with unreliability $u = 1 - \bar{r}$, the correct threshold is $u > 1 - \tau = 0.34$, not $u > 0.66$. The implementation also exposes minimum and hybrid gate modes for RGA-v2 experiments. These variants are reported as diagnostic candidates, not as established improvements: in the current k -of- D run, the configured minimum threshold $\tau_{\min} = 0.34$ is below the observed minimum reliability, so the minimum branch does not fire and the hybrid gate collapses to the mean-gate behavior. All reported base-RGA tables use the original mean gate unless noted. This makes RGA different from confidence weighting. Confidence weighting uses the local score confidence directly. RGA uses validation calibration, test-time drift evidence, and sharpness to decide when a domain should be trusted less.

4) *Theory Boundary*: The companion thesis appendix replaces the earlier bounded-drift proposition with three claims used by this manuscript. First, global KS references are confounded under category mixture shift: if $P_{\text{val}} = \sum_c \pi_c P_c$ and $P_{\text{test}} = \sum_c \pi'_c P_c$, then $\text{KS}(P_{\text{val}}, P_{\text{test}})$ can be positive even when every category distribution P_c is unchanged. Category-conditioned KS removes that population-level false signal under pure mixture shift. Second, the current mean gate has a dilution boundary. If k of D domains collapse to reliability

zero and the others remain at one, the gate fires only when $k > D(1 - \tau)$. For $D = 4, \tau = 0.66$, one failed domain can be hidden while two or more failed domains can activate the path. Third, switching is beneficial only when

$$\pi q_1 \Delta_1 > (1 - \pi) q_0 \Delta_0,$$

where π is degradation prevalence, q_1 is gate power, q_0 is false activation rate, Δ_1 is degraded-regime benefit, and Δ_0 is clean false-switching cost. The paper’s empirical claims are bounded by these measurable quantities rather than by a universal “RGA is better” theorem.

5) *Validation-Selected RGA+ Head*: The implementation also reports a separate RGA+ candidate rather than silently replacing the base RGA number. RGA+ has two parts. First, a reliability-boosted fusion head trains candidate classifiers on the fusion train fold using flattened domain evidence, missingness indicators, score and confidence summary statistics, and RGA reliability summaries. The current candidate set contains histogram gradient-boosted heads and calibrated logistic heads. The selected head is the one with the best validation ROC-AUC; test labels are never used for model or hyperparameter selection. Second, a validation-only router can choose among static attention, base RGA, RGA+ boosted fusion, and the strong score-level baselines. The selected candidate is written into the result artifact so that a baseline selection is not relabeled as a reliability-gate gain. When the validation fold has only one class, the router defaults to base RGA because the validation labels cannot rank candidates.

RGA+ is therefore a supervised fusion extension of the reliability system, not a claim that the original gate alone dominates all baselines. This distinction is central to the MVTec results: base RGA remains the diagnostic mechanism; RGA+ is the validation-selected modeling head used when a two-class fusion split exists.

6) *Counterfactual Domain Attribution*: For sample i , let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \quad (6)$$